

Chapter 20 - Numerical Integration

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#Numerical Integration

The numerical computation of the definite integral when it becomes very difficult or infeasible or impossible to find its exact value. In other words, numerical integration is concerned with the computation of an approximate value for the definite integral rather than finding an exact value.

Definite Integral and Area Computation

The fundamental theorem of integral calculus states that

$$\int_a^b f(x) dx = F(b) - F(a)$$

which gives the signed area bounded by the curve $f(x)$, the vertical lines $x = a$ and $x = b$ and the x-axis. This provides an exact value for the definite integral. But there are some functions which do not have an elementary antiderivative. Such integrals can still be computed appropriately using numerical integration.

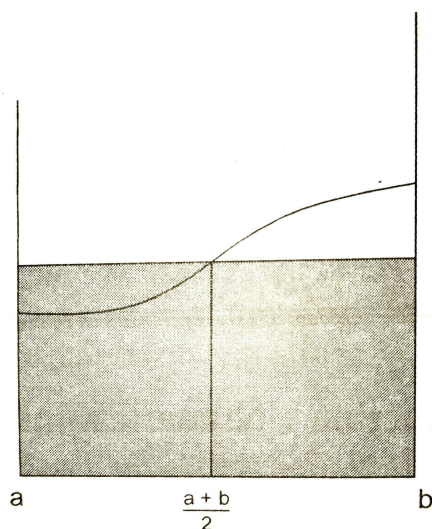
Rectangle Rule or Mid-point Rule

The rectangle rule (or mid-point rule) is an elementary method of numerical integration to approximate the definite integral using the rectangle.

According to rectangle rule,

$$\int_a^b f(x) dx \approx (b - a) f\left(\frac{a + b}{2}\right)$$

Here, The constant function $f\left(\frac{a + b}{2}\right)$ is known as an interpolating function or approximation to function f .



#Trapezium Rule

The trapezium (or trapezoidal) rule is a numerical integration method used to approximate the area under a curve by dividing it into small trapeziums.

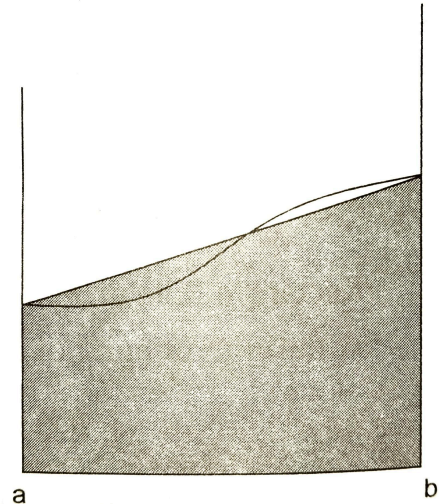
Elementary Trapezium Rule

The elementary trapezium rule approximates the definite integral of a function by using a single trapezium over the entire interval $[a, b]$.

According to elementary trapezium rule,

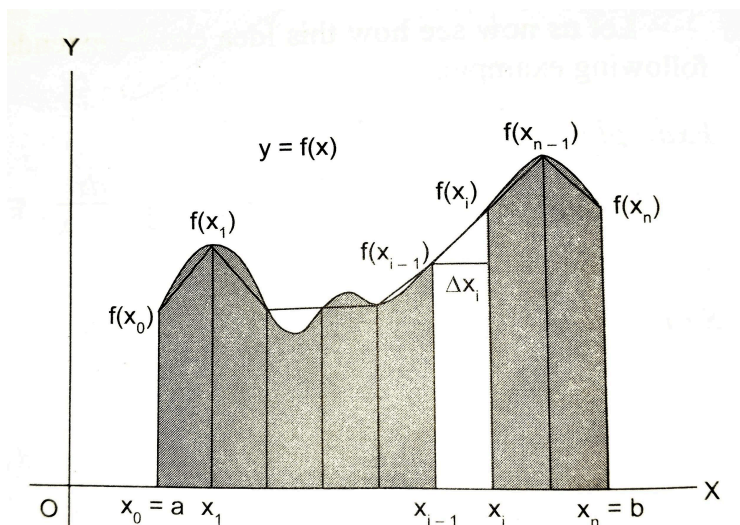
$$\int_a^b f(x) dx \approx (b - a) \frac{f(a) + f(b)}{2}$$

Here, the interpolating function is an affine function (a polynomial of degree 1) passing through the points $(a, f(a))$ and $(b, f(b))$.



Composite Trapezium Rule

The composite trapezium rule is a numerical integration technique used to approximate the area under a curve by dividing the total interval into multiple small subintervals or strips.



According to composite trapezium rule,

$$\int_a^b f(x) dx \approx \frac{h}{2} (y_0 + 2y_1 + 2y_2 + \dots + 2y_{n-1} + y_n) \text{ where } h = \frac{b-a}{n}$$

Estimation of Error

$$\text{Error(Trapezium Rule)} = \frac{(b-a)^3 M}{12n^2}$$

where M is the largest value of $|f''(x)|$ in the interval $[a, b]$.

#Simpson's Rule

Simpson's rule is a numerical integration method that approximates the area under a curve by fitting curve arcs (higher polynomials) to the function's points.

Simpson's 1/3 Rule

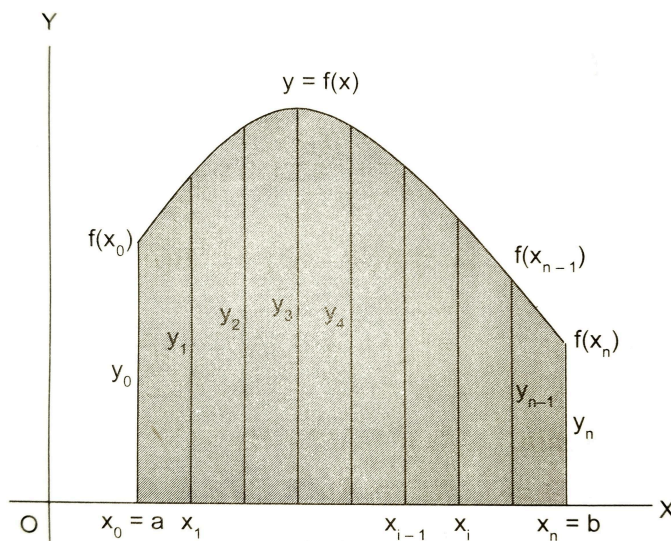
Simpson's 1/3 rule is a method for numerical integration that fits quadratic polynomials (second-degree) through sets of three points.

According to Simpson's rule,

$$\int_a^b f(x) dx \approx \frac{h}{3} [f(a) + 4f(\frac{a+b}{2}) + f(b)] \text{ where } h = \frac{b-a}{2}$$

Composite Simpson's 1/3 Rule

The composite Simpson's 1/3 rule is used to approximate the integral of a function over a large interval by breaking it into n smaller subintervals (where n is a multiple of 2) and applying Simpson's 1/3 rule to each pair of subintervals.



According to Simpson's 1/3 rule,

$$\int_a^b f(x) dx \approx \frac{h}{3} (y_0 + 4y_1 + 2y_2 + 4y_3 + 2y_4 + \dots + 2y_{n-2} + 4y_{n-1} + y_n) \text{ where } h = \frac{b-a}{n}$$

Estimation of Error

$$\text{Error(Simpson's } \frac{1}{3} \text{ Rule)} = \frac{(b-a)^5 M}{180n^4}$$

where M is the largest value of $|f^{(4)}(x)|$ in the interval $[a, b]$.

Simpson's 3/8 rule

Simpson's 3/8 rule is a method for numerical integration that fits cubic polynomials (third-degree) through sets of four points.

According to Simpson's 3/8 rule,

$$\int_a^b f(x) dx \approx \frac{3h}{8} (y_0 + 3y_1 + 3y_2 + y_3) \text{ where } h = \frac{b-a}{3}$$

Composite Simpson's 3/8 rule

The composite Simpson's 3/8 rule is used to approximate the integral of a function over a large interval by breaking it into n smaller subintervals (where n is a multiple of 3) and applying Simpson's 3/8 rule to each pair of subintervals.

According to Simpson's 3/8 rule,

$$\int_a^b f(x) dx \approx \frac{3h}{8} (y_0 + 3y_1 + 2y_2 + 3y_3 + 2y_4 + \dots + 3y_{n-2} + 2y_{n-1} + y_n) \text{ where } h = \frac{b-a}{n}$$

Estimation of Error

$$\text{Error(Simpson's } \frac{3}{8} \text{ Rule)} = \frac{(b-a)^5 M}{80n^4}$$

where M is the largest value of $|f^{(4)}(x)|$ in the interval $[a, b]$.